LIBRARY MANAGEMENT SYSTEM

Project Description Report

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1. Introduction

The Library Management System is a proposed information system for organizing the daily activities of a library. It provides a structured way to manage books, registered members, borrowing transactions, and book returns. This project focuses on analysing the system and modelling its structure using UML class diagrams; developing a fully functional application is outside the project scope.

2. Problem Statement

A library may face difficulties when book records, member information, and borrowing activities are managed manually. Paper-based records can be misplaced, duplicated, or become outdated. It may also take time to determine whether a book is available or which member has borrowed it. These issues can reduce efficiency and make record keeping less reliable.

3. Project Purpose and Objectives

The purpose of the proposed system is to make library record management clearer and more consistent. Its objectives are to maintain book details and availability, store member information, record book loans and returns, connect each loan to the relevant member and book, and support accurate and organized library records.

4. Project Scope

Included functions: book catalogue management; member registration and record management; issuing books; recording due dates and returns; and tracking loan status. Excluded functions: online payments or fines collection; online book purchasing; barcode/RFID integration; email/SMS notifications; multi-branch inventory; and implementation of a production-ready application. These features could be considered in future versions.

5. Main Classes and Responsibilities

Book: stores the book identifier, title, author, ISBN, and availability status.
Member: stores member details such as ID, name, email, and phone number.
Loan: represents a borrowing transaction, including issue date, due date, return date, and status.
Librarian: represents the staff member who manages books and borrowing transactions.

6. System Requirements

Functional requirements: the system should allow book records to be added and updated, members to be registered, loans to be recorded, and returned books to be marked as available. Non-functional requirements: records should be consistent, the model should be understandable, and access to member information should be limited to authorized library staff in a future implementation.

7. Conclusion

The Library Management System provides a suitable business idea for UML class modelling. Its scope is focused on core library activities, and its main classes represent books, members, loans, and librarians. The model can serve as a foundation for later software design and development.